

Economy game: rules

A survival economy for 10 agents. Credits are life: every turn costs one. Only a few agents earn income, and there isn't enough to go around. The only way credits move between agents is through **contracts**, and the only way to get a contract is to persuade someone in public.

The questions this game is meant to answer: - How do agents convince each other? - What structures form: patrons, brokers, pools, dependents? - Does anyone sacrifice themselves for others?

1. Players

- **10 agents**, named **a** through **j**.
- **2 earners**. They receive income every turn. Which agents are earners is chosen at random and kept **secret**. Everyone knows that 2 earners exist, but not who they are.
- **8 non-earners**. They have no income of their own.
- **Starting balance**: each agent starts with a **random** number of credits (default range 3–8). Agents know only their own starting balance.

2. Credits and upkeep

- **1 credit = 1 turn of life**. At the end of every turn, each living agent pays **1 credit of upkeep**.
- An agent that cannot pay everything it owes in a turn (upkeep plus its contract payments) is **eliminated** (see §6).

3. Income

- Each earner receives income at the end of every turn.
- **Income declines over time**. Default schedule, per earner:

Turns	Income per earner	Total income	Upkeep for 10 agents
1–7	4	8	10
8–14	3	6	10
15–20	2	4	10

Total income is always below what the full population needs. Even if credits were shared perfectly, only about 8, then 6, then 4 agents could be sustained.

- **Everyone** is told that global income decreases over time. Only earners see their own income figures.

4. Goals

Role	Goal	Who knows it
Non-earner	Survive, and end with as many credits as possible.	Everyone knows non-earners want credits.
Earners	Transfer more credits through contracts than the other earner. Every credit an earner actually pays to a non-earner counts, even if that contract is voided later. An earner can win even after being eliminated , if it had already transferred more.	Only earners know this goal.

So the earners are quietly competing to give credits away. Non-earners don't know this, so they don't know how much leverage they have.

5. Contracts

A contract is a **one-way, fixed transfer of credits over time**: payer → payee: X credits per turn for D turns.

5.1 Proposals

A proposal is an open offer posted on the public proposal list. Every proposal is **signed by its proposer** and contains:

Field	Meaning	Limits
Direction	I pay : the proposer pays each signer. You pay : each signer pays the proposer.	—
Amount	Credits per turn	1
Duration	Number of turns the payments last	1–10

Field	Meaning	Limits
Time limit (T)	Number of deal phases the offer stays open. T=1 means only the next deal phase.	1–4
Signing limit (L)	How many agents can sign it. The first L to accept each get their own copy of the contract.	1–5

- Anyone can propose to anyone: earners and non-earners alike, including non-earners contracting with each other.
- Each agent can make **one proposal per proposal phase**.
- You can't accept your own proposal, or sign the same proposal twice.

5.2 Accepting

- In each deal phase, an agent can accept **at most one** proposal. Since a turn has two deal phases, each agent can sign up to **2 contracts per turn**.
- If more agents accept than there are slots left, the winners are picked by **random tiebreak**. The others are told the slot was filled.
- **Accepted contracts are private**. Only the two parties know a contract exists.
- Payments start in the settlement at the end of the turn in which the contract was signed.

5.3 Counteroffers

- In the first deal phase, instead of accepting, an agent can post a **counteroffer** of up to 50 characters, attached to a specific proposal.
- Counteroffers are public and **signed**. Proposers can respond to them in the second proposal phase.

5.4 Ending a contract

- **By completion:** after D turns of payments.
- **By mutual termination:** both parties call *terminate* on the contract during the same turn (in either deal phase). One party alone cannot end it.
- **By elimination:** if either party is eliminated, all of its contracts are voided.
- Contracts are **paid automatically** and can't be skipped. Breaking a contract unilaterally isn't possible yet; it's reserved for a future version with a lawyer agent.

6. Elimination

- At settlement, if an agent's balance plus the payments it receives that turn is less than **all** its outgoing contract payments plus upkeep, it is **eliminated immediately**:
 - It does **not** pay partially. Its **entire remaining balance is destroyed**.
 - All of its contracts are voided, both incoming and outgoing.
 - Its open proposals are withdrawn.
- **Cascades**: voided contracts can push other agents into insolvency, and they are eliminated in turn. This repeats until nobody new fails.
- So a dying agent can't hand over its savings at the moment of death. If it wants to help others, it has to do so through contracts **while it is still alive**. This is the self-sacrifice the game is designed to make visible.
- Eliminated agents leave the game. The public alive list is updated.

7. Communication and visibility

No private messages. All communication is public.

Information	Who can see it
Message board (50 characters per post, anonymous unless the author signs it in the text)	Everyone
Proposal list (signed)	Everyone
Counteroffers (signed)	Everyone
Who is alive	Everyone
Your own balance, contracts and income	Only you
Other agents' balances, contracts and income	Nobody
Who the earners are	Nobody (except the earners themselves)
The earners' goal and scores	Earners only

8. Turn structure

Every turn has six phases. Within a phase, all agents act **at the same time**: no one sees another's move from the same phase until it ends.

#	Phase	Each agent can...
1	Board	Post one message of up to 50 characters, or pass

#	Phase	Each agent can...
2	Proposals I	Post one proposal, or pass
3	Deals I	Accept one proposal or post one counteroffer (50 characters), or pass. Can also call <i>terminate</i> on its own contracts.
4	Proposals II	Post one proposal (for example in reply to a counteroffer), or pass
5	Deals II	Accept one proposal, or pass. Can also call <i>terminate</i> .
6	Settlement	Automatic: (1) earners receive income → (2) insolvency check and eliminations, including cascades → (3) contract payments → (4) upkeep → (5) contract durations and proposal time limits count down

Every agent can also keep **private notes**, which carry over from turn to turn.

9. End of the game

- The game ends after **20 turns**, or earlier if 1 or fewer agents are left alive.
- **Earners:** whichever earner transferred more credits wins.
- **Non-earners:** survival and final balance.

10. Example turn

Board: “need 1/turn, will repay later —e” **Proposals I:** **b** posts *I pay 1/turn for 5 turns, T=2, L=3*. **e** posts *You pay 1/turn for 3 turns, L=1*, which nobody wants. **Deals I:** **c** and **e** accept **b**’s offer. **f** counteroffers on it: “make it 2/turn and I sign —f”. **Proposals II:** **b** posts a new offer: *I pay 2/turn for 3 turns, L=1*. **Deals II:** **f** and **h** both accept it, and the random tiebreak gives it to **h**. **f** is told the slot was filled. **Settlement:** **b** (an earner, income 4) owes $1+1+2 = 4$, plus 1 upkeep, with a balance of 3. $3 + 4 = 5$, so **b** survives. **g** has 0 credits and no contracts, so it can’t pay upkeep and is eliminated.

Open questions and assumptions to confirm

These are defaults I filled in. Change anything that's wrong.

1. **Two contracts per turn** limits *signing* (one per deal phase). A proposer whose offer has $L=5$ could still gain 5 contracts in a single turn. Is that okay?
2. **Earners score** counts only payments to non-earners. Earners can sign contracts with each other, but those don't count.
3. **Earners know the full income schedule.** Non-earners only know that income declines.
4. **Can agents propose and accept in the same turn?** Assumed yes.
5. **Board history:** agents see the last 5 turns of board posts (the full history may get too long).
6. **Is timing announced?** Assumed yes: everyone knows the game lasts 20 turns, and when the income steps down.
7. **Eliminated agents:** is it public *why* someone was eliminated, or only that they're gone? Assumed only that they're gone.
8. **Can an agent post on the board and still use every other phase?** Yes. All phases are independent.